

Recap!

Given a function $f: A \rightarrow B$

1. f is onto iff

$$[\forall b \in B: \exists a \in A: f(a) = b]$$

2. f is 1:1 iff

$$[\forall a_1, a_2 \in A: a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)]$$

$\equiv$

$$[\forall a_1, a_2 \in A: f(a_1) = f(a_2) \Rightarrow a_1 = a_2]$$

Let $f: A \rightarrow B$ (f is a function)

$f: A \rightarrow B$ is onto $\Rightarrow |A| \geq |B|$

$f: A \rightarrow B$ is 1:1 $\Rightarrow |A| \leq |B|$

$f: A \rightarrow B$ is a bijection $\Rightarrow |A| = |B|$

Theorem 9.13

Pigeon Hole Principle (PHP)

Let A and B be sets with a function $f: A \rightarrow B$

If $|A| > |B|$, then there exist 2 distinct $a_1, a_2 \in A$ s.t. $f(a_1) = f(a_2)$

Let's try to write this using a quantified expression.

$$|A| > |B| \Rightarrow [\exists a_1, a_2 \in A: f(a_1) = f(a_2)]$$

Is there a similarity with this theorem and def of 1:1?

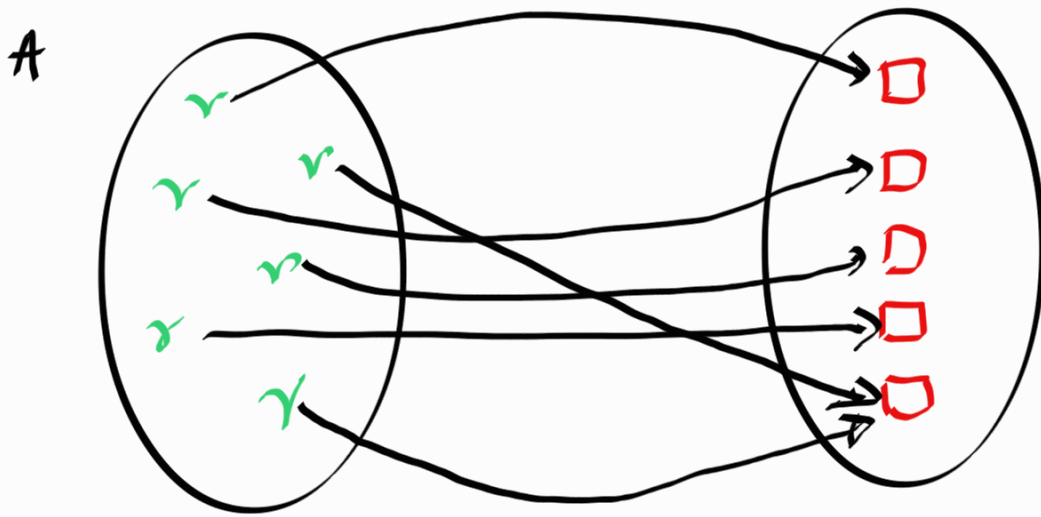
$$f: A \rightarrow B \text{ is 1:1} \Rightarrow |A| \leq |B|$$

$$|A| > |B| \Rightarrow f: A \rightarrow B \text{ is not 1:1}$$

PCP with pigeons:

Suppose you have a set A of pigeons
where $|A| = n+1$

Also, you have a set of cubbies (B)
where $|B| = n$



Claim: Among 13 people, at least
2 of them share the same birth month.

Proof:

Let A (pigeons) be the set of ¹³ people.

Let B (cubbies) be the set of 12 months.

Let $f: A \rightarrow B$, $f(a) = a$'s birth month.

first, we need to prove that f is
a function.

1. $\forall a \in A: f(a)$ is defined

This is true, because every person has a birth month.

2. $\forall a \in A: f(a)$ is unique.

True because every person can only be born in one month.

3. $\forall a \in A: f(a) \in B$.

True, because birth month has to be one of the 12 months.

$\therefore f$ is a function.

Now, we have 2 sets A, B where
 $f: A \rightarrow B$ and $|A| > |B|$ ($13 > 12$)

Hence, $[\exists a_1, a_2 \in A: (a_1 \neq a_2) \wedge f(a_1) = f(a_2)]$

according to PHP.

This means there exists two distinct people who share the same birth month.